

Traversal Pattern (Level 3)

1. Second Largest Element

```
arr = [10,22,111,33,555,2222]

largest = float('-inf')
second = float('-inf')

for i in arr:
    if i > largest:
        second = largest
        largest = i

    elif i > second and i != largest:
        second = i

print(second)
```

2. Second Smallest Element

```
arr = [100,22,111,33,555,2222]

smallest = float('inf')
second = float('inf')

for i in arr:
    if i < smallest:
        second = smallest
        smallest = i

    elif i < second and i != smallest:
        second = i

print(second)
```

3. Maximum Difference

```
arr = [10,22,111,33,555,2222]

maximum = max(arr)
minimum = min(arr)

max_difference = maximum - minimum

print(max_difference)
```

4. Find Missing Number (Simple)

```
arr = [1, 2, 3, 4, 6]

n = max(arr)

expected = n * (n+1) // 2

actual = sum(arr)

print(expected - actual)
```

5. Remove Duplicates (Basic)

```
arr = [111,22,111,33,555,2222]

result = []

for i in arr:
    if i not in result:
        result.append(i)

print(result)
```

6. Maximum Consecutive Ones

```
arr = [1, 1, 0, 1, 1, 1,1]
count = 0
maximum = 0

for i in arr:
    if i == 1:
        count = count + 1
        maximum = max(maximum, count)
    else:
        count = 0

print(count)
```

7. Majority Element (Brute Force)

```
arr = [111, 111, 111, 33, 555, 111]

for i in arr:
    count = 0

    for j in arr:
        if j == i:
            count = count + 1

    if count > len(arr) // 2:
        print(i)
```

8. Product of Array

```
arr = [10,22,111,33,555,2222]

product = 1

for i in arr:
    product = product * i

print(product)
```

9. Find Pivot Index (Basic)

```
arr = [4, 6, 99, 3, 7]

for i in range(len(arr)):

    left = sum(arr[:i])
    right = sum(arr[i+1:])

    if left == right:
        print(i)
```

10. Best Time to Buy and Sell Stock (Traversal)

```
prices = [7, 1, 5, 3, 6, 4]

minimum = float('inf')

profit = 0

for price in prices:

    minimum = min(minimum, price)

    profit = max(profit, price - minimum)

print(profit)
```